Tencent

The Tencent Advertising Algorithm Challenge 2026: Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation

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We present the KDD Cup 2026 Tencent Advertising Algorithm Challenge *Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation*, a large-scale competition and benchmark for unified recommendation modeling. The competition focuses on post-click conversion rate (pCVR) prediction. For each user–ad ranking instance, participants observe two heterogeneous sources of information: (i) non-sequential multi-field features covering user, ad, context, and pre-computed cross representations, and (ii) chronologically ordered user behavior histories from multiple behavioral domains, where each behavior is accompanied by heterogeneous side information such as item attributes, behavior/scenario indicators, and timestamps. The central challenge is to transform these heterogeneous signals into compatible representations and jointly process them with a *homogeneous, stackable backbone*, enabling *effective cross-source interactions* and *hardware-efficient scaling with high GPU utilization*. Submitted models predict the conversion probability of the target ad and are evaluated by AUC under controlled computational resources and strict inference constraints.

The competition consisted of academic and industry tracks conducted over two online rounds. Submissions were evaluated by AUC on separate test sets under controlled computational resources and a *strict no-ensemble policy*. All teams trained and evaluated their models on the organizer-provided AngelML platform using large-scale Tencent Advertising logs and standardized access to high-performance GPUs.

The challenge attracted 13,913 registered participants forming 5,746 teams from 52 countries and regions and 907 universities, research institutes, and companies. Fifty academic-track teams and twenty industry-track teams advanced to Round 2. We subsequently received 46 solution papers, of which 23 were accepted for presentation at the KDD Cup workshop. We release de-identified public samples, detailed feature statistics, and starter resources at this link[a] to support further research on unified recommendation modeling.

[a] https://huggingface.co/TAAC2026

1 Introduction

Recommender systems are foundational to both *content recommendation* (e.g., feeds, short videos, news, e-commerce) and *advertising* (e.g., CTR/CVR prediction and auction-based ranking), shaping user experience, platform engagement, and revenue. At today's scale, these systems operate under massive traffic and strict latency constraints, making billions of decisions daily. For example, Weixin and WeChat have over one billion monthly active users globally, generating massive volumes of personalized recommendation requests in real time [42]. Meanwhile, global digital advertising spending reached approximately $790 billion in 2024 [22], underscoring the economic importance of large-scale advertising systems. Together, these statistics highlight the broad social and economic impact of advances in recommender and advertising systems.

Over the past two decades, recommendation research has evolved from collaborative filtering and linear predictors

to deep representation learning and large-scale ranking architectures, broadly progressing along two major branches: (i) *feature interaction models*, which model dependencies among multi-field categorical and contextual features, and (ii) *sequential models*, which capture users' dynamically evolving interests from their historical behaviors. Feature interaction modeling has progressively evolved from *shallow explicit interaction* to *deep interaction modeling* and, more recently, *large-scale model scaling*. Early factorization-based methods in the FM family efficiently model second-order interactions through inner products between field embeddings [36, 20, 30, 41]. Subsequent DNN-based architectures, ranging from Wide & Deep and DeepFM to NFM, AutoInt, xDeepFM, and DCN-V2, substantially enhance representation capacity by combining explicit feature crosses with increasingly expressive implicit or high-order interaction mechanisms [7, 15, 13, 24, 40, 46]. More recently, as industrial recommenders move toward larger model capacity, scaling-oriented approaches such as Heterogeneous-MoE, Multi-Embedding, DHEN, Wukong, RankMixer, RankUp, RankElastor, and UniFormer shift the focus from designing individual feature crosses toward increasing the usable representation and interaction capacity of the overall model, while addressing challenges in representation dimensionality, model depth, parameter scaling, and hardware efficiency [31, 14, 53, 54, 62, 5, 23, 4].

In parallel, sequential recommendation has evolved from modeling relatively short behavior histories to capturing increasingly long and heterogeneous user trajectories. Early and short-sequence models, including SASRec and the DIN family, develop self-attention, target-aware attention, interest evolution, and temporal interaction mechanisms to extract user interests from recent behavior sequences [21, 59, 58, 11, 60]. As industrial platforms accumulate much longer behavior histories, long- and lifelong-sequence models such as SIM, TWIN/TWIN-V2, LONGER, DARE, and MUSE increasingly rely on memory, retrieval, sampling, sequence compression, representation decoupling, and efficient attention mechanisms to extend the modeling horizon while controlling computational and serving costs [33, 35, 1, 3, 39, 2, 10, 50, 16, 55].

Towards Unified Modeling. Despite the substantial progress of both branches, feature interaction modeling and sequential recommendation have historically evolved as largely separate architectural components. Traditional behavior-aware CTR and ranking models commonly follow a modular *encode-then-interact* paradigm: a dedicated interest or sequence-modeling module first compresses user behavior histories into one or a few fixed-length representations, which are subsequently concatenated with non-sequential user, item, and contextual features and processed by downstream MLP or feature-interaction networks [59, 58, 11, 6, 35, 28, 38, 60]. This separation remains prevalent in more recent industrial systems, where increasingly powerful sequence encoders are coupled with separately designed cross, MoE, or heterogeneous-expert interaction backbones and fused at a later stage [12, 51, 31, 29, 63, 26]. While such modular designs are effective and allow the two components to evolve independently, they often compress behavioral information before rich cross-source interaction, limiting repeated bidirectional conditioning between individual behaviors and non-sequential features and making model scaling dependent on coordinating heterogeneous subnetworks. Motivated by the growing importance of jointly exploiting these heterogeneous signals, recent studies, including InterFormer, OneTrans, HyFormer, EST, MixFormer, HeMix, TokenFormer, and UniFormer, have explored increasingly integrated architectures for jointly modeling sequential and non-sequential information [52, 57, 18, 27, 17, 45, 61, 4].

Nevertheless, unified recommendation modeling remains far from settled. Existing approaches differ substantially in their tokenization strategies, interaction topologies, sequence compression mechanisms, degree of architectural homogeneity, and scaling strategies. Moreover, their effectiveness is typically demonstrated on different datasets and under different feature spaces, sequence lengths, computational budgets, and serving constraints, making direct comparison difficult. Consequently, it remains unclear which architectural principles are broadly effective, how well different unified designs scale with model depth and capacity, and whether their accuracy gains can be retained under realistic GPU and inference constraints.

These open questions motivate a common benchmark for systematically studying unified recommendation modeling under controlled industrial-scale settings. We therefore organized the KDD Cup 2026 Tencent Advertising Algorithm Challenge *Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation*, providing a shared task that jointly exposes heterogeneous multi-field features and multi-domain behavioral sequences under standardized data, computational, and inference constraints. As illustrated in Figure 1, the challenge particularly encourages participants to explore effective heterogeneous tokenization, homogeneous and stackable unified blocks, fine-grained interactions between sequential and non-sequential signals, and hardware-efficient model scaling.

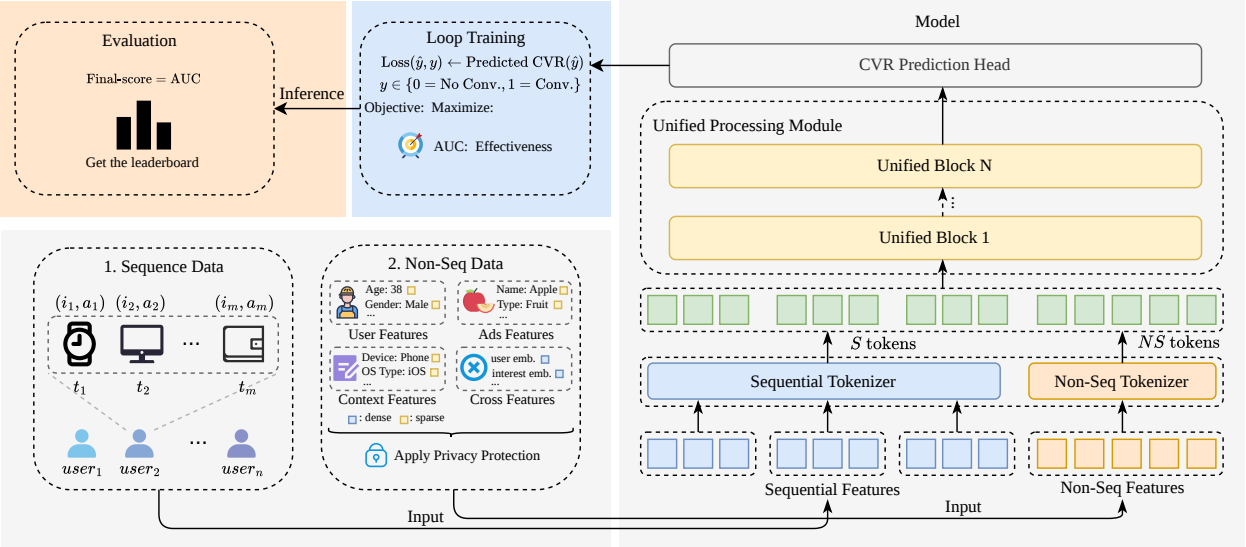


Figure 1: Overview of the competition framework. **Bottom-left (Input):** Each instance consists of (1) sequential behavior data, i.e., chronological interaction histories for individual users, with associated items and timestamps, and (2) non-sequential multi-field features grouped into four categories: User Features, Ads Features, Context Features (sparse), and Cross Features (dense embeddings), all subject to privacy protection. **Right (Model):** Participants are encouraged to transform sequential and non-sequential inputs into compatible representations and jointly process them through a stack of unified blocks within a homogeneous backbone, followed by a CVR prediction head. The illustrated tokenization represents a general design paradigm rather than a prescribed model architecture. **Top-left (Training and Evaluation):** The model is trained with binary supervision using cross-entropy loss, potentially together with auxiliary objectives, and submissions are ranked by AUC on the official test set.

2 Challenge Setting

Problem setting. In industrial recommendation and advertising ranking, each prediction instance jointly depends on two heterogeneous sources of information: (i) rich *non-sequential multi-field features*, including user, target-ad, contextual, dense, and pre-computed cross features, and (ii) *multi-domain user behavior histories*, where each historical behavior is associated with heterogeneous side information such as item attributes, behavior type, scenario, and timestamp. Conventional systems commonly process these two sources with specialized feature-interaction and sequence-modeling modules and combine their representations only after largely separate processing. Although effective, such modular designs restrict repeated fine-grained interaction between sequential and non-sequential signals and make it difficult to construct homogeneous, deeply stackable architectures. Accordingly, the challenge encourages participants to develop unified and scalable models that jointly exploit both sources under realistic computational and inference constraints.

Data and notation. Each training or test example corresponds to a user u, a request/context c, and a target ad i. The model observes the following inputs:

- **Non-sequential multi-field features.** We denote the collection of non-sequential features by

$$\mathcal{F}_{\text{ns}} = \left\{ f^{(1)}, \ldots, f^{(K)} \right\}, \tag{1}$$

  covering user/profile features, contextual features, target-ad features, and other sparse or dense representations. Depending on the field type, $f^{(k)}$ may be a categorical value, a multi-value feature, or a dense vector.

- **Multi-domain behavior histories.** User behaviors are organized into D behavioral domains,

$$\mathcal{S}_u = \left\{ S_u^{(1)}, \ldots, S_u^{(D)} \right\}, \qquad S_u^{(d)} = \left(x_1^{(d)}, \ldots, x_{T_d}^{(d)} \right), \tag{2}$$

where T_d is the history length of domain d. Each behavior event contains multiple side-information fields:

$$x_t^{(d)} = \left(g_t^{(d,1)}, \ldots, g_t^{(d,M_d)} \right), \tag{3}$$

representing information such as the interacted item's attributes, behavior type, scenario, and timestamp.

Prediction objective. Let Φ_θ denote a participant model that jointly processes the heterogeneous non-sequential features, multi-domain behavior histories, and target-ad information. The predicted conversion score is

$$s_\theta(u, c, i) = \Phi_\theta \left(\mathcal{F}_{\text{ns}}, \mathcal{S}_u, i \right). \tag{4}$$

where request/context information c is included in $\mathcal{F}_{\text{ns}}$. Given labeled examples $\{(u_n, c_n, i_n, y_n)\}_{n=1}^N$, where $y_n \in \{0, 1\}$ indicates whether a post-click conversion occurs, participants aim to learn Φ_θ such that the resulting scores achieve high ranking quality, measured by AUC on the hidden evaluation set, while satisfying the prescribed computational and inference constraints.

Unified modeling. The competition does not prescribe a specific tokenization or interaction architecture. Participants may independently design how heterogeneous fields, behavior histories, and target information are represented, compressed, and jointly modeled. As illustrated in Figure 1, the challenge particularly encourages architectures that map these heterogeneous inputs into compatible representations and repeatedly process them through *homogeneous and stackable unified blocks*. This formulation allows diverse designs—including field tokens, behavior tokens, query or intent tokens, sequence summaries, and other learned representations—while retaining the central objective of effective cross-source interaction and hardware-efficient scaling.

2.1 Competition Protocol

Tracks and eligibility. The competition consisted of two parallel tracks with the same prediction task but independent leaderboards and track-specific inference constraints. The **Academic Track** was reserved for teams whose members were all current students at universities or colleges, with cross-institution teams permitted. The **Industry Track** was open to other eligible participants, including practitioners from research institutes and companies, and adopted a stricter inference-efficiency requirement. Participation was team-based: each team consisted of **1 to 3 members**, and each participant could belong to only one team.

Competition procedure. The competition comprised **two online rounds**, followed by solution-paper review and presentation at the KDD Cup workshop. Round 1 was open to all eligible teams and emphasized broad architectural exploration. The top-50 teams in the Academic Track and top-20 teams in the Industry Track advanced to Round 2, where participants were evaluated under substantially larger data and computational scales. After Round 2, finalist teams submitted technical reports describing their solutions, and accepted works were presented at the KDD Cup workshop.

Submission, evaluation, and fairness. Across both rounds, teams could make at most **three submissions per day**. Submissions were evaluated server-side on hidden test sets, with AUC serving as the leaderboard metric among models satisfying the corresponding track-specific inference-time constraints. To emphasize single-model architectural innovation and ensure fair comparison, **model ensembling was prohibited**. Throughout both rounds, the organizing committee conducted multiple rounds of *AI-assisted code review*, supplemented by manual verification, to assess compliance with the competition rules. The review pipeline analyzed submitted code for prohibited ensemble methods and potential plagiarism, including suspicious structural and semantic similarities across submissions. Several teams were found to have employed ensemble strategies, most commonly simple or weighted averaging of predictions from multiple models. These submissions were deemed non-compliant, and the corresponding results were subsequently invalidated. To the best of our knowledge, TAAC 2026 is the first large-scale public recommendation competition to report the use of AI-assisted code review, complemented by manual verification, to enforce an explicit no-ensemble policy. This human–AI review pipeline enabled scalable inspection of a large number of submissions while preserving human judgment for final compliance decisions, helping maintain a fair comparison of single-model architectural innovations.

Round 1: broad exploration. Round 1 provided a moderate-scale setting for rapid model development and architectural exploration. Depending on the track, the training data contained roughly **one to two million** interaction instances, together with several hundred thousand inference instances and approximately **120 feature columns** spanning user, item, contextual, dense, and multi-domain sequential signals. Each team was allocated a fractional high-performance GPU slot on the AngelML platform, allowing a large number of teams to experiment under a standardized computational environment.

Round 2: large-scale evaluation. Round 2 substantially increased both data and model-development capacity to emphasize scalable recommendation modeling. The datasets expanded by more than an order of magnitude to **tens of millions of training instances** and **millions of inference instances**, while the feature schema grew to approximately **140 columns** with richer sparse and dense signals. Each finalist team received **7 high-performance GPUs** on AngelML, enabling the exploration of deeper, wider, and more computationally demanding unified architectures under the same controlled evaluation protocol.

2.2 Awards

The competition offered a total prize pool of **5.9 million RMB**, with **3.6 million RMB** allocated to the Academic Track and **1.7 million RMB** to the Industry Track. Both tracks awarded prizes to the highest-ranked teams, with additional honorable mentions for top-performing teams in the Academic Track.

Beyond leaderboard performance, the competition established two **Technical Innovation Awards**, each worth **300,000 RMB**, to recognize contributions that advanced the design of large-scale unified recommender systems. The awards focused on (i) *scaling laws for unified recommendation models*, and (ii) *unified block design* for bridging sequential modeling and multi-field feature interaction. These awards were intended to recognize architectural and scaling innovations beyond incremental accuracy improvements. They were not disclosed in advance and were announced at the KDD Cup workshop based on the technical contributions demonstrated by the finalist solutions.

2.3 KDD Cup Workshop

Following Round 2, finalist teams were invited to submit solution papers through OpenReview[1]. We received **46 submissions**, of which **23** were accepted for presentation, including **8 oral presentations** and **15 posters**. The workshop brought together competition finalists and invited researchers from academia and industry to discuss winning solutions, scalable unified recommendation architectures, and emerging directions revealed by the competition.

3 Data

The competition data consist of de-identified Tencent Ads interaction logs and are designed to support the central problem of this challenge: jointly modeling user behavior sequences and non-sequential multi-field features within a unified backbone. Each prediction instance corresponds to a user–item ranking opportunity with a response label, a target timestamp, static or slowly changing user/item fields, and multi-domain historical behavior sequences. The public data resources include both first-round and second-round samples; both use flat Parquet files to expose typed feature columns for model development.[2]

Dataset construction. The competition dataset was constructed from attributed pCVR advertising logs from Weixin Channels. To improve the representativeness and difficulty of the prediction task, we removed all samples associated with conversion types exhibiting high conversion rates, as well as samples involving dual optimization objectives or non-optimization objectives.

We adopted a temporally ordered split, with the test window occurring strictly after the training period. Users were kept disjoint between the training and test sets to prevent user-level information leakage. To reduce temporal sampling bias, test instances were sampled proportionally across all 24 hours of the day. Likewise, the training set was proportionally sampled from multiple days, preserving the temporal coverage of the original logs.

[1] https://openreview.net/group?id=KDD.org/2026/Cup/UniRec

[2] First-round sample: https://huggingface.co/datasets/TAAC2026/data_sample_1000. Second-round sample: https://huggingface.co/datasets/TAAC2026/second_round_sample_1000.

Category	First-round Count	Second-round Count	Type	Description
ID & Label	5	5	int64/int32	Core identifiers, label type, label time, and timestamp.
User Int Features	46	54	int64/ list<int64>	Discrete user features, including scalar profile attributes and multi-value preference or attribute arrays.
User Dense Features	10	17	list<float>	Continuous user-side signals, including embedding features and dense values aligned with selected user integer arrays.
Item Int Features	14	17	int64/ list<int64>	Discrete item features such as item categories, types, and multi-label item attributes.
Item Dense Features	0	4	list<float>	Continuous item-side embedding features, available in the second-round sample.
Domain Sequence Features	45	45	list<int64>	Historical behavior sequences from four domains, exposing multi-domain temporal signals.

Table 1: Column groups in the TAAC2026 public first-round and second-round data samples.

Privacy-preserving release. All released records are de-identified before publication. Personally identifiable information, raw creative content, and business-sensitive raw attributes are excluded. Sparse identifiers and categorical values are represented as anonymized integer IDs, while selected continuous signals are provided as pre-computed dense feature arrays. Participants therefore observe typed feature columns suitable for modeling, but do not receive raw text, images, videos, URLs, or direct user/ad identifiers. The public samples are released under the CC BY-NC 4.0 license for non-commercial research and educational use.

Feature schema. Table 1 summarizes the public sample schemas for both rounds. The first-round sample contains 120 columns, while the second-round sample extends the same flat-column design to 142 columns. The five ID and label columns are `user_id`, `item_id`, `label_type`, `label_time`, and `timestamp`. User-side sparse features include scalar columns such as `user_int_feats_1` and multi-value columns such as `user_int_feats_62`. Some user dense columns are standalone embeddings, while others are aligned with corresponding integer-array columns; for example, `user_dense_feats_62` has the same element order as `user_int_feats_62`. Item-side features follow the same principle, with scalar or multi-value integer columns such as `item_int_feats_5`; the second-round sample additionally includes item dense embedding columns such as `item_dense_feats_124`. The sequence portion is organized into four behavioral domains in both samples: `domain_a_seq_{38-46}`, `domain_b_seq_{67-79,88}`, `domain_c_seq_{27-37,47}`, and `domain_d_seq_{17-26}`.

Dataset scale. The first-round competition contains both academic-track and industry-track interaction datasets. In the academic track, the training split contains 1,010,000 interaction samples stored in Parquet files with 120 columns, while the inference split contains 310,000 samples under the same schema. In the industry track, the training and inference splits contain 2,099,956 and 523,244 samples, respectively, and are also represented with 120 columns. The second-round competition is substantially larger and uses an expanded 142-column schema. Its academic-track training split contains 21,995,548 interaction samples, and the corresponding inference split contains 6,098,994 samples. The second-round industry track further scales the data to 34,822,423 training samples and 12,251,082 inference samples. Across both rounds, each record combines user–item identifiers, label and timestamp fields, user-side and item-side sparse/dense features, and multi-domain sequential behavior features. We report aggregated feature column-type coverage statistics for the first- and second-round datasets in Appendix Tables 2 and 3, respectively.

4 Baseline and Evaluation

4.1 Baseline

We use HyFormer as the reference baseline for joint sequence modeling and feature interaction [18]. HyFormer represents the behavior history as a sequence and the non-sequential fields as feature tokens, and connects the two inputs through Query Generation, Query Decoding, and Query Boosting.

Let $\{F_k\}_{k=1}^{K}$ denote the non-sequential feature tokens, where $F_k \in \mathbb{R}^{1\times D}$, and let $E(S_u)$ denote the behavior-sequence representation. HyFormer constructs the global information and initial queries as

$$\mathbf{g} = \mathrm{Concat}\left(F_1, \ldots, F_K, \mathrm{MeanPool}(E(S_u))\right), \tag{5}$$

$$Q^{(0)} = \left[\mathrm{FFN}_1(\mathbf{g}), \ldots, \mathrm{FFN}_{N_q}(\mathbf{g})\right] \in \mathbb{R}^{N_q\times D}. \tag{6}$$

The initial queries are reused and updated in subsequent HyFormer layers.

For sequence encoding, HyFormer supports Transformer, LONGER-style cross-attention, and SwiGLU. At layer l, the sequence representation $H^{(l)}$ is projected into key and value representations, followed by Query Decoding and Query Boosting:

$$\begin{aligned} K^{(l)} &= H^{(l)}W_K^{(l)}, \qquad V^{(l)} = H^{(l)}W_V^{(l)}, \\ \widehat{Q}^{(l)} &= \mathrm{CrossAttn}\big(Q^{(l-1)}, K^{(l)}, V^{(l)}\big), \\ \widetilde{Q}^{(l)} &= \mathrm{QueryBoost}\big(\mathrm{Concat}\big(\widehat{Q}^{(l)}, F_1, \ldots, F_K\big)\big). \end{aligned} \tag{7}$$

The boosted queries $\widetilde{Q}^{(l)}$ are used as the input queries of the next HyFormer layer.

The output of the top HyFormer layer is passed to downstream MLPs to estimate the response probability. The baseline is trained with binary cross-entropy:

$$\widehat{y}_\theta(u, c, i) = \sigma\big(s_\theta(u, c, i)\big), \tag{8}$$

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{|\mathcal{D}|}\sum_{(u,c,i,S_u,y)\in\mathcal{D}} \left[y\log\widehat{y}_\theta(u, c, i) + (1-y)\log\big(1-\widehat{y}_\theta(u, c, i)\big)\right]. \tag{9}$$

4.2 Evaluation Metrics

We use the area under the receiver operating characteristic curve (AUC) as the official leaderboard metric. For each test instance n, let $s_n = s_\theta(u_n, c_n, i_n) \in \mathbb{R}$ denote the prediction score and let $y_n \in \{0, 1\}$ denote the corresponding response label. Define $\mathcal{P} = \{n : y_n = 1\}$ and $\mathcal{N} = \{n : y_n = 0\}$ as the positive and negative index sets, respectively. AUC is computed as

$$\mathrm{AUC} = \frac{1}{|\mathcal{P}||\mathcal{N}|}\sum_{p\in\mathcal{P}}\sum_{n\in\mathcal{N}}\left[\mathbb{I}(s_p > s_n) + \frac{1}{2}\mathbb{I}(s_p = s_n)\right]. \tag{10}$$

Here, $\mathbb{I}(\cdot)$ is the indicator function. Equation (10) measures the probability that a positive instance receives a higher score than a negative instance, with ties counted as one-half. The official leaderboard is ordered by AUC on the hidden test set.

4.3 Evaluation Protocol

Evaluation was conducted in a controlled server-side environment on the official test set. Test labels remained confidential. Participants submitted prediction files or containerized inference code, and the official evaluation service computed AUC without exposing the labels.

4.4 Competition Platform

We hosted the challenge on a Tencent-operated official competition website and ran all training, validation, and submission evaluation through our in-house **AngelML** platform, providing a uniform and reproducible environment for all teams. The website served as the single source of truth for registration, rules, data documentation, baselines, announcements, and leaderboard status. Importantly, the platform is **accessible to participants worldwide**, ensuring

that **teams from all regions** can register, train, and submit under the same standardized setting. AngelML provides standardized software stacks (containers/environments), centralized job scheduling, and consistent hardware access, so that participants do not need to provision their own compute and can reproduce results under identical configurations.

To ensure accessibility and fairness, we provisioned GPU resources directly on AngelML across both rounds. During the first round, each team was allocated a virtual slot equivalent to one-fifth of a high-performance GPU to support iterative development and online submissions; during the second round, each finalist team received **7 GPUs** to enable stronger training and thorough ablations under a controlled environment. We also released a complete starter kit, including data loaders, preprocessing scripts, evaluation code, and a strong baseline implementation, so that participants can reproduce an end-to-end pipeline from day one. Finally, we enforced strict isolation and security policies (e.g., sandboxed evaluation, controlled I/O, and standardized dependency management) to minimize leakage risks and maintain a stable and fair competition environment.

5 Challenge Summary

5.1 Registration

The Tencent Advertising Algorithm Challenge 2026 attracted broad participation from both academia and industry worldwide. Participants represented universities, research institutes, and major technology companies, reflecting strong interest in unified recommendation modeling across both research and practitioner communities.

Registration scale and geographic coverage. In total, 13,913 participants registered and formed 5,746 teams. Based on the affiliation information provided during registration, participants represented 52 countries and regions across six continents and were affiliated with 907 universities, research institutes, and companies worldwide. These statistics reflect the broad geographic and institutional reach of the challenge and the strong interest in unified recommendation modeling across both research and practitioner communities.

Track and industry participation. Among the 5,746 registered teams, 3,799 teams selected the Academic Track, accounting for 66.1% of all teams, while 1,947 teams selected the Industry Track, accounting for 33.9%. The Industry Track attracted practitioners affiliated with many major technology companies. The most represented domestic companies included Meituan, Kuaishou, Alibaba, Baidu, Ant Group, ByteDance, Huawei, Xiaomi, and JD.com, while international participants reported affiliations with companies such as VK, Microsoft, Meta, Amazon, Shopee, NVIDIA, Google, Tesla, and Qualcomm. This broad industrial representation suggests that the challenge addressed problems of direct relevance to large-scale recommendation and advertising systems.

Academic institutional participation. Academic participation was similarly broad. In mainland China, the institutions with the largest numbers of registrants included the University of Science and Technology of China, Tsinghua University, Shanghai Jiao Tong University, and Fudan University. Strong participation was also observed from universities in Hong Kong, particularly the Chinese University of Hong Kong, City University of Hong Kong, and the Hong Kong University of Science and Technology. Outside Greater China, highly represented institutions included the National University of Singapore, the University of California, Los Angeles, Nanyang Technological University, Carnegie Mellon University, and the University of Sydney. Compared with the previous edition, the number of overseas registrants increased by 110%, indicating a substantial expansion in the competition's international reach.

5.2 Solution Overview

We summarize the solutions of the three highest-ranked teams in each track, as well as the two innovation-award-winning teams. See the corresponding workshop papers for further details.

Academic Track: First Place. The winning team proposed CRAFT, which handles sparse categorical fields, dense vectors, contextual features, and multi-domain behavior histories through type-aware tokenization and a shared stackable backbone. Its central design replaces interaction-centric cross-layer propagation with contextual feature transport: non-sequential features are reliability-weighted to produce sample-specific control signals that regulate how intent and sequence representations evolve across depth.

For each behavior domain, CRAFT generates intent tokens from non-sequential context and pooled behavioral evidence. Within every block, the contextual transport module predicts channel-wise scaling, bias, and bounded memory signals for intent and sequence states, producing a context-conditioned representation displacement while preserving the initial information path. Sequence encoders then refine behavioral dynamics, intent tokens retrieve sequence evidence through cross-attention, and a token-subspace mixer performs local field refinement. The final ranking score is computed from the resulting intent representations rather than directly from the original feature tokens.

This design provides a unified interface between heterogeneous context and behavioral evidence without relying on late fusion or independent feature and sequence towers. The distilled CRAFT model achieved a test AUC of 0.838090, surpassing the interaction-based distilled baseline at 0.837983. Stacking CRAFT to six blocks further achieved 0.838148, while a later $d_{\text{model}} = 480$ configuration reached a post-submission test AUC of 0.838192.

Academic Track: Second Place. The runner-up team proposed the UniDAC architecture for large-scale CVR prediction. The model is motivated by three characteristics of industrial ranking data: strong non-sequential features can dominate sequential signals, user behaviors are fragmented across multiple domains, and naive parameter scaling often brings limited gains. Building on a HyFormer-style query-based framework [18], UniDAC introduces a Dual-stack Encoding scheme to strengthen sequential modeling. A LONGER branch [2] compresses old histories into a summary token while preserving recent events to capture long-range interest evolution, whereas a parallel attention-free SwiGLU branch [37] focuses on local event-level patterns. UniDAC further introduces a Cross-domain Interaction module, which summarizes each behavior domain into a domain-level token, exchanges complementary information through cross-domain attention, and injects the resulting context back into each sequence via gated residual connections. Candidate-aware query generation and RankMixer-style token mixing [62] are used to improve the interaction between target-conditioned sequential evidence and non-sequential features, while a SoftMoE prediction head [34] expands the output-side modeling capacity. For efficient scaling, the model controls high-cardinality sparse embeddings through fixed-budget hashing and ID dropout and adopts FlashAttention-2 [8] to reduce the cost of long-sequence attention. Together, these scaling-friendly architectural and system optimizations enabled the model to scale from 100M to 1B parameters, achieving an AUC of 0.83792 and winning second place in the Academic Track.

Academic Track: Third Place. The third-place team proposed **UniTokenFormer**, a unified token-enhanced framework for post-click conversion rate (pCVR) prediction built on HyFormer [18]. Instead of treating tokenization as a fixed preprocessing step, the method systematically designs token construction, target-aware enhancement, and capacity allocation. Heterogeneous user-side and item-side fields are first concatenated and projected before being split into multiple non-sequential tokens, preserving cross-field interactions as token capacity increases. Candidate-item tokens further condition multi-domain behavior sequences and generate DIN-style interest tokens [59]. User, item, and interest tokens are fused before domain-specific queries decode the behavior sequences with LENS-enhanced target-conditioned gates and positional biases [48], while domain auxiliary objectives strengthen per-sequence supervision. The method further treats token allocation as an explicit scaling dimension. The study showed that, under comparable dense model sizes, increasing item-side tokens was more effective than increasing user-side tokens, while an item-heavy joint allocation enabled a single-block model to scale to 1.063 billion dense parameters. Combined with field-wise feature preprocessing, bidirectional time-gap encoding, grouped sparse/dense optimization, exponential moving averages, BF16 training, and graph compilation, the approach achieved consistent gains over the official HyFormer baseline.

Industry Track: First Place. The winning team proposed **QueryFormer**, a unified post-click conversion rate (pCVR) model for jointly modeling heterogeneous non-sequential fields and multi-domain behavioral sequences. Building on recent progress in unified modeling [27], the model replaces projection-only query construction with explicit token-to-query attention. Its stackable field–sequence block first refines user and item field tokens with self- and cross-attention. Item tokens then retrieve evidence from four behavior domains, while behavior-initialized probes cross-attend to the refined field context to generate queries for subsequent sequence interaction. To scale capacity under strict inference constraints, QueryFormer organizes tokenization as an embedding matrix with H independent columns. These views are evaluated in parallel, and their sequence queries are packed into shared-parameter cross-attention to amortize key/value projections and GPU kernel launches. Within the team's experimental grid, increasing H consistently improved validation AUC: the $H{=}4$ configuration gained 0.00066 AUC over $H{=}1$, while $H{=}8$ required only $1.89\times$ the per-sample inference latency of $H{=}1$. Replacing the cross-attention query generator with an MLP caused the largest

individual ablation drop, supporting query generation as the main architectural contribution. QueryFormer achieved first place in the Industry Track with an official test AUC of 0.83254.

Industry Track: Second Place. The runner-up team proposed **UniDot**, whose central component is the FM Highway, a dedicated path that carries explicit dot-product feature interactions around the residual stack and delivers them directly to the classifier [25]. Rather than computing second-order interactions only once at the input, as in FM [36] and its deep variants, the model emits a factorization-machine term at every layer from that layer's current token representations, so interactions are re-derived as the representations evolve in depth. This places the design alongside a line of recent industrial recommenders that reintroduce an explicit dot-product matching step inside an otherwise attention-based backbone [47], but routes those scores past the trunk entirely, so explicit low-order signals are not diluted by repeated nonlinear mixing and the classifier retains direct access to interaction evidence at every level of abstraction. The highway allows the rest of the network to form a single homogeneous stack: because explicit interactions are guaranteed a route to the output, the trunk is free to specialize in implicit mixing. The trunk stacks a repeated macro-block in which a token-mixing bus and a sequence-retrieval bus (item tokens cross-attending to the behavior histories) operate in parallel over a shared token space. The design is motivated by the observation that the embedding inner product of collaborative filtering and the query–key score of dot-product attention [44] share the same primitive, while the two buses exchange state at every layer through an MLP-Mixer [43] fusion. The sequence side is embedded once per forward pass and shared by all consumers, bounding inference latency [47]. Training uses a dual sparse/dense optimizer (Adagrad [9] on the embedding tables and Muon [19] on the dense parameters), an auxiliary conversion-delay head, and multi-path mutual learning [56] across the model's parallel prediction paths.

Industry Track: Third Place. The third-place team proposed **UniBridge**, a stackable *decode-then-mix* architecture for jointly modeling behavioral sequences and multi-field features. UniBridge first applies lightweight local temporal modeling to each behavior stream. Queries conditioned on the candidate item then attend to the behavior history and compress it into a small set of intent tokens. This design reduces the cost of long-sequence modeling and avoids applying global self-attention to the full behavior history [2]. The resulting behavior tokens are combined with user and item field tokens in a shared residual stream. In addition, a parameter-free RankMixer rewire [62] is used to model their interactions. Compared with fully unified Transformer architectures [57], UniBridge keeps the number of mixed tokens small and fixed. This design supports early interaction between sequence and field features with low computational overhead.

Winner of the Unified Block Innovation Award. The awarded team proposed MaskRec, a topology-masked unified architecture for jointly modeling heterogeneous feature interactions and multi-domain user behavior sequences. Building on recent efforts to unify feature interaction and sequence modeling in industrial recommender systems [18, 57, 61], MaskRec transforms heterogeneous non-sequential features, contextual signals, multi-domain behavior sequences, and auxiliary representations into a shared token space and models them with a unified Transformer-style backbone. Its core component, TopoMask, extends the masking mechanism of Transformer attention [44] from temporal constraints to topology-aware interaction constraints, explicitly regulating token-level query–key connections according to token roles and behavior-domain relationships. Global memory tokens and domain-level memory tokens are further introduced as learnable aggregation nodes for global and domain-specific information. To inject candidate-conditioned interaction signals before unified modeling, MaskRec further introduces DualQ, which performs bidirectional cross-attention between user-side and candidate-item-side tokens, followed by asymmetric attentive pooling and domain-conditioned feature-wise linear modulation (FiLM) [32] to generate domain-aware query tokens. The resulting design enables source-local modeling, domain-level aggregation, and controlled cross-source interaction within the same backbone and consistently outperforms the HyFormer-style baseline on the competition dataset.

Winner of the Scaling Law Innovation Award. The awarded team developed CoSFormer, a single-model architecture for post-click conversion prediction built on the competition's HyFormer-style baseline [18]. Its central contribution is information-bearing token co-scaling. In the RankMixer-style backbone [62], model width factorizes as $d_{\text{model}} = T d_{\text{sub}}$, where T is the number of query and context tokens. Rather than widening an unchanged representation, CoSFormer fixes $d_{\text{sub}} = 4$, appends tokens derived from new predictive signals, and widens the shared backbone in lockstep. The resulting 48-token stream unifies static context and four behavior sequences through independently projected pretrained vectors, explicit user–item crosses, TIN-based target-time readouts [60], and multi-resolution history summaries. A

MaskNet-inspired dual stream [49], token-type-private feed-forward networks, and candidate-aware attention further process these heterogeneous channels.

To isolate the scaling gain, the team evaluated a nine-step semantic-token ladder against two counterfactuals: widening the existing bank and appending learned empty tokens at a fixed per-token subspace. Width-only scaling degraded performance from the first scaling step, while empty tokens produced short-lived gains before turning downward. Semantic co-scaling instead improved AUC until later token families became redundant, providing an empirical stopping rule. The results showed that useful scaling came from growing predictive information and model capacity together, rather than from token count or parameter count alone. Interestingly, this finding independently aligns with the scaling philosophy of RankUp [5], which also expands model capacity by constructing richer and more informative input representations rather than simply widening hidden layers or embedding dimensions. Together, these independent results suggest that effective scaling in recommendation may depend not only on making models *wider*, but also on making the representations being scaled *more informative*.

6 Conclusion

This paper presents the KDD Cup 2026 Tencent Advertising Algorithm Challenge, a large-scale benchmark for jointly modeling heterogeneous multi-field features and multi-domain user behavior sequences under controlled data, computational, and inference constraints. Through two competition tracks and two progressively scaled rounds, the challenge provides a common setting for systematically evaluating unified recommendation architectures and studying their effectiveness, stackability, scalability, and hardware efficiency. The competition results reveal several recurring design principles. Successful solutions increasingly move beyond conventional late-fusion pipelines by introducing richer tokenization, target- or intent-aware cross-source interaction, and stackable mechanisms for jointly processing sequential and non-sequential signals. Meanwhile, the strongest scaling-oriented solutions suggest that increasing model capacity is most effective when accompanied by correspondingly richer and more informative representations, rather than by naively increasing hidden dimensions or parameter counts alone.

Together, these findings highlight that the next stage of large-scale recommendation modeling requires the co-design of *representation, interaction, scaling, and system efficiency*. We hope that the released de-identified data samples, feature statistics, starter resources, winning solutions, and competition findings will provide a useful foundation for future research on unified and scalable recommender systems.

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A The Complete Feature Coverage Tables

Table 2: Feature column-type coverage of the first-round competition datasets. Mean, minimum, and maximum coverage values are reported in percent. The `Full` column counts features with at least 99% coverage. Rows with zero columns denote feature types absent from the first-round schema.

Dataset	Feature type	#Cols	Mean	Min	Max	Full ($\geq 99\%$)
Academic train	User Integer	46	70.96	7.96	99.99	7 (15%)
	Item Integer	14	77.61	12.05	100.00	10 (71%)
	User Dense	10	90.59	51.99	99.87	2 (20%)
	Item Dense	0	–	–	–	–
	Domain-A Sequence	9	99.93	99.93	99.93	9 (100%)
	Domain-B Sequence	14	99.89	99.89	99.89	14 (100%)
	Domain-C Sequence	12	99.95	99.95	99.95	12 (100%)
	Domain-D Sequence	10	99.90	99.90	99.90	10 (100%)
	All features	**115**	**84.80**	**7.96**	**100.00**	**64 (56%)**
Academic inference	User Integer	46	71.17	8.37	100.00	7 (15%)
	Item Integer	14	77.43	12.38	100.00	10 (71%)
	User Dense	10	91.22	51.69	99.89	2 (20%)
	Item Dense	0	–	–	–	–
	Domain-A Sequence	9	99.97	99.97	99.97	9 (100%)
	Domain-B Sequence	14	99.92	99.92	99.92	14 (100%)
	Domain-C Sequence	12	99.97	99.97	99.97	12 (100%)
	Domain-D Sequence	10	99.94	99.94	99.94	10 (100%)
	All features	**115**	**84.94**	**8.37**	**100.00**	**64 (56%)**
Industry train	User Integer	46	70.99	8.33	99.99	7 (15%)
	Item Integer	14	77.21	11.30	99.93	10 (71%)
	User Dense	10	90.60	50.85	99.87	2 (20%)
	Item Dense	0	–	–	–	–
	Domain-A Sequence	9	99.83	99.83	99.83	9 (100%)
	Domain-B Sequence	14	99.87	99.87	99.87	14 (100%)
	Domain-C Sequence	12	99.97	99.97	99.97	12 (100%)
	Domain-D Sequence	10	99.90	99.90	99.90	10 (100%)
	All features	**115**	**84.76**	**8.33**	**99.99**	**64 (56%)**
Industry inference	User Integer	46	71.17	8.82	100.00	7 (15%)
	Item Integer	14	77.20	10.75	99.91	10 (71%)
	User Dense	10	90.92	51.50	99.86	2 (20%)
	Item Dense	0	–	–	–	–
	Domain-A Sequence	9	99.53	99.53	99.53	9 (100%)
	Domain-B Sequence	14	99.40	99.40	99.40	14 (100%)
	Domain-C Sequence	12	99.97	99.97	99.97	12 (100%)
	Domain-D Sequence	10	95.69	95.69	95.69	0 (0%)
	All features	**115**	**84.41**	**8.82**	**100.00**	**54 (47%)**

Table 3: Feature column-type coverage of the second-round competition datasets. Mean, minimum, and maximum coverage values are reported in percent. The `Full` column counts features with at least 99% coverage.

Dataset	Feature type	#Cols	Mean	Min	Max	Full ($\geq$ 99%)
Academic train	User Integer	54	74.77	8.63	99.96	11 (20%)
	Item Integer	17	81.48	13.13	99.97	13 (76%)
	User Dense	17	93.34	53.00	99.84	8 (47%)
	Item Dense	4	97.04	93.25	99.18	1 (25%)
	Domain-A Sequence	9	99.61	99.61	99.61	9 (100%)
	Domain-B Sequence	14	97.80	97.80	97.80	0 (0%)
	Domain-C Sequence	12	99.93	99.93	99.93	12 (100%)
	Domain-D Sequence	10	81.78	81.78	81.78	0 (0%)
	All features	**137**	**85.26**	**8.63**	**99.97**	**54 (39%)**
Academic inference	User Integer	54	75.25	8.60	99.95	11 (20%)
	Item Integer	17	81.76	15.26	99.93	13 (76%)
	User Dense	17	94.40	53.27	99.88	8 (47%)
	Item Dense	4	96.88	93.39	99.04	1 (25%)
	Domain-A Sequence	9	99.54	99.54	99.54	9 (100%)
	Domain-B Sequence	14	97.99	97.99	97.99	0 (0%)
	Domain-C Sequence	12	99.96	99.96	99.96	12 (100%)
	Domain-D Sequence	10	81.15	81.15	81.15	0 (0%)
	All features	**137**	**85.58**	**8.60**	**99.96**	**54 (39%)**
Industry train	User Integer	54	74.17	8.67	99.96	11 (20%)
	Item Integer	17	81.21	12.35	99.97	13 (76%)
	User Dense	17	92.80	51.41	99.85	7 (41%)
	Item Dense	4	96.06	92.72	99.11	2 (50%)
	Domain-A Sequence	9	99.79	99.79	99.79	9 (100%)
	Domain-B Sequence	14	99.13	99.13	99.13	14 (100%)
	Domain-C Sequence	12	99.92	99.92	99.92	12 (100%)
	Domain-D Sequence	10	96.54	96.54	96.54	0 (0%)
	All features	**137**	**86.12**	**8.67**	**99.97**	**68 (50%)**
Industry inference	User Integer	54	75.01	8.51	99.96	11 (20%)
	Item Integer	17	81.97	15.76	99.97	13 (76%)
	User Dense	17	94.20	51.45	99.89	8 (47%)
	Item Dense	4	94.01	83.75	99.16	2 (50%)
	Domain-A Sequence	9	99.79	99.79	99.79	9 (100%)
	Domain-B Sequence	14	99.48	99.48	99.48	14 (100%)
	Domain-C Sequence	12	99.95	99.95	99.95	12 (100%)
	Domain-D Sequence	10	98.62	98.62	98.62	0 (0%)
	All features	**137**	**86.84**	**8.51**	**99.97**	**69 (50%)**